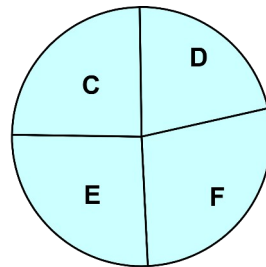


SCHOOL OF MEDICAL LABORATORY TECHNOLOGY
EPIDEMIOLOGY (SMLT 212) FINAL EXAMINATION
Calendar Year 2008/09

Date: April, ___/ 2009
Time: 2.00 hrs

For Questions 1-21 and Q27-30, unless indicated Please choose the best answer and circle it.

1. In the definition of epidemiology, which term of the following does NOT exist?
 - a. distribution
 - b. determinant
 - c. application
 - d. disease
 - e. non of the above
2. Descriptive study tries to answer the following EXCEPT:
 - a. Who
 - b. where
 - c. why
 - d. when
 - e. non of the above
3. Which of the following is NOT primary use of epidemiology?
 - a. individual decision
 - b. improve career
 - c. Identify population at risk
 - d. community health assessment
 - e. non of the above
4. If a particular disease is caused by any of the following sufficient causes diagrammed below, which components if any are **NECESSARY** cause?
(Circle ALL that apply.)
 - a. A
 - b. B
 - c. C
 - d. D
 - e. E
 - f. F



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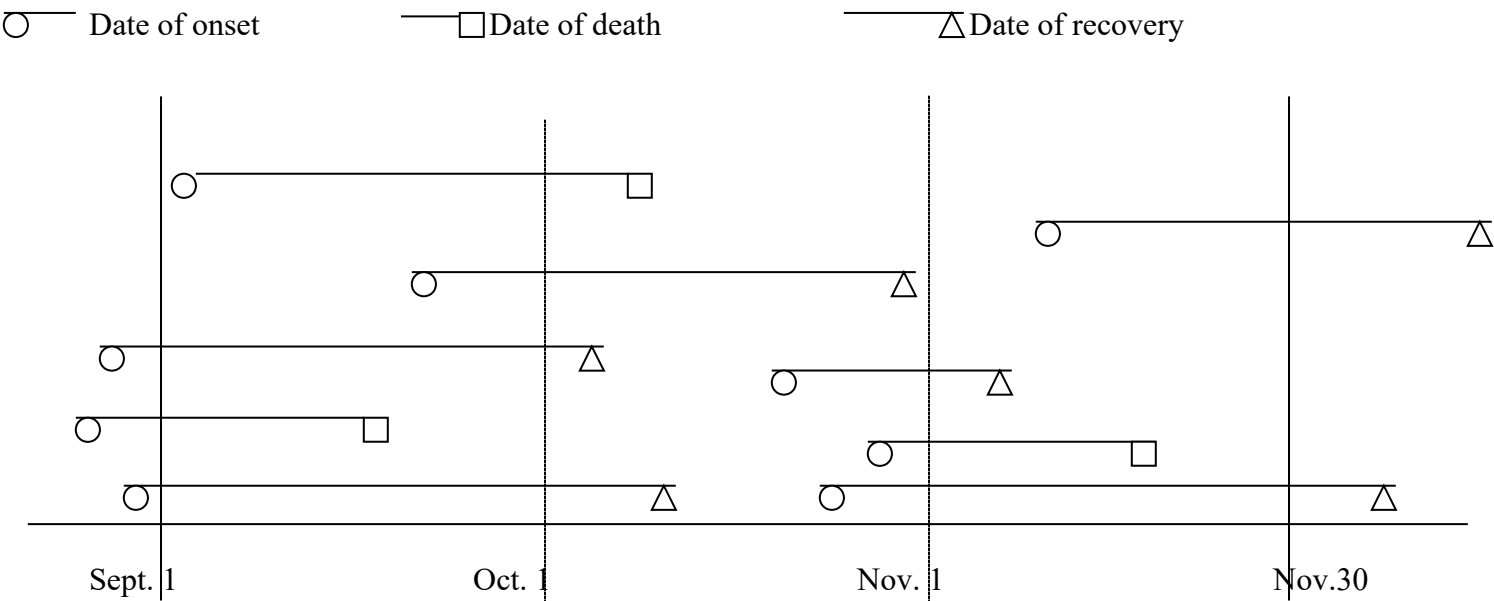
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14. Which of the following illnesses has soil as a reservoir?
- a. Clostridium tetanus b. Malaria d. HIV/ AIDS
 d. Rabies e. non of the above
15. Which of the following conditions or terms describes well about **PERSON**?
- a. age b. urban c. country
 d. a & b are correct e. b & c are correct
16. Measurements of data in epidemiology include **ALL, EXCEPT**?
- a. Count b. ratio c. proportion
 d. population e. rate

To answer Questions 17-21 use the following figure showing episodes of an illness X in a population of 100 visited for 3 consecutive months.



17. What is the incidence of illness X among the population in the month Sept 1 to Oct 1?
- a. 20 /1000 popn. b. 20.6 / 1000 popn. c. 50/ 1000 popn.
 d. 51.5/ 1000 popn. e. non of the above
18. What is the prevalence of illness X among the population on Oct. 1 ?
- a. 40 /1000 popn. b. 40.2 / 1000 popn. c. 40.4/ 1000 popn.
 d. 50/ 1000 popn. e. non of the above
19. What is the prevalence of illness X among the population from Sept. 1 to Nov. 30 ?-----
- a. 91.4 /1000 popn. b. 90 / 1000 popn. c. 95.7/ 1000 popn.

- d. 100/ 1000 popn. e. non of the above

20. What is the mid year population of the population between Sept. 1 and Nov. 30?

- a. 100 people b. 91 people c. 93 people
 d. 98.5 people e. Non of the above

21. What is the denominator for the incidence illness X among the population from Sept. 1 to Nov 30?

- a. 6 b. 9 c. 3
 d. 7 e. Non of the above

For questions 22 to 26 use the following 2 x 2 table of new screening test.

		True disease (golden standard test)		Total
		Positive	Negative	
Screening test	Positive	480	120	600
	Negative	20	880	900
Total		500	1000	1500

22. Specificity _____

23. Sensitivity _____

24. Predictive value positive _____

25. Predictive value negative _____

26. Prevalence _____ (Assume that the population screened has been well sampled)

27. Which of the following studies is **NOT** a Descriptive Study design?

- a. Correlational b. Case report / Case series c. Cross-sectional surveys
d. Cohort study e. non of the above

28. The fraction shown below is a: (Circle all that apply)

$$\frac{\text{\# of women in Ethiopia who died from malaria in 1991}}{\text{\# of women in Ethiopia who died from cervical cancer in 1991}}$$

- a. Ratio b. Proportion c. Mortality rate
 d. incidence rate e. rate

29. One of the following is cause for bias?

- a. Sampling error b. Accurate measurement c. Precise measurement
 d. Reliable measurement e. Non of the above

30. Inaccuracy can be prevented through all the following strategies, **EXCEPT** :
- a. Training of data collectors
 - b. Blinding about specimens
 - c. Checking instruments
 - d. Following standardized guidelines
 - e. None of the above

Answer “True” for statements correctly stated and “False” for incorrectly stated sentences.

1. All ratios are proportions but all proportions are not ratios. ____f____
2. As prevalence of a disease increases its predictive value negative decreases. ____t____
3. Cross sectional study is also called incidence study. ____f____
4. Descriptive studies are used to allocate resources efficiently and to plan effective prevention or education programs. ____t____
5. Ecological study uses data from the entire population to compare disease frequencies. ____t____
6. Sporadic spread– is an epidemic, which occurs worldwide. E.g. influenza, cholera, AIDS. __f__
7. Propagated epidemic: occur as a result of the exposure of a group of persons or population to a common source for the etiological agent, such as a contaminated water supply, or the food in a certain restaurant. ____f____
8. A disease is said to be pandemic in a population or area if it is usually present there at a more or less stable level. ____f____
9. A carrier is a person without apparent disease, but who are able to transmit an agent of a disease. ____t____
10. Tests performed in persons with symptoms with a symptom or a sign of an illness are usually termed as diagnostic tests. ____t____

Give the Alphabet that gives the best answer in the space provided of column “A” by matching from column “B”.

A

B

- | | | |
|-------------|----------------------------|--|
| _____d_____ | 1. Acute disease | a. Primary or true cause |
| _____e_____ | 2. Non infectious disease | b. Malaria |
| _____a_____ | 3. Agent of a disease | c. Place |
| _____f_____ | 4. Necessary Cause | d. Rapid onset and short duration |
| _____g_____ | 5. Reservoir | e. Caused by other than living organisms |
| _____h_____ | 6. Portal of entry for TBC | f. Cause with out the agent the disease doesn't occur |
| _____ | 7. Having animal reservoir | g. Habitat in which an agent lives, develops and multiblies. |
| _____i_____ | 8. Tertiary prevention | h. Respiratory inhalation |
| _____c_____ | 9. Geographic variation | i. Rehabilitation |
| _____j_____ | 10. Natality | j. Birth |